

PROJECT

5.4

**LAUNCHING VPC
RESOURCES**
NETWORKING SERIES

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Objective

This project aims to launch EC2 instances in public and private subnets within an AWS VPC. The public instance supports internet-facing tasks, while the private one is used for secure backend operations.

To simplify the setup, the Amazon VPC wizard is used to quickly create the VPC with subnets and route tables, following best practices for speed and security.

Services

Amazon EC2 - Provides scalable virtual servers to run applications in the cloud.

Amazon VPC - Creates an isolated network environment for securely launching AWS resources.

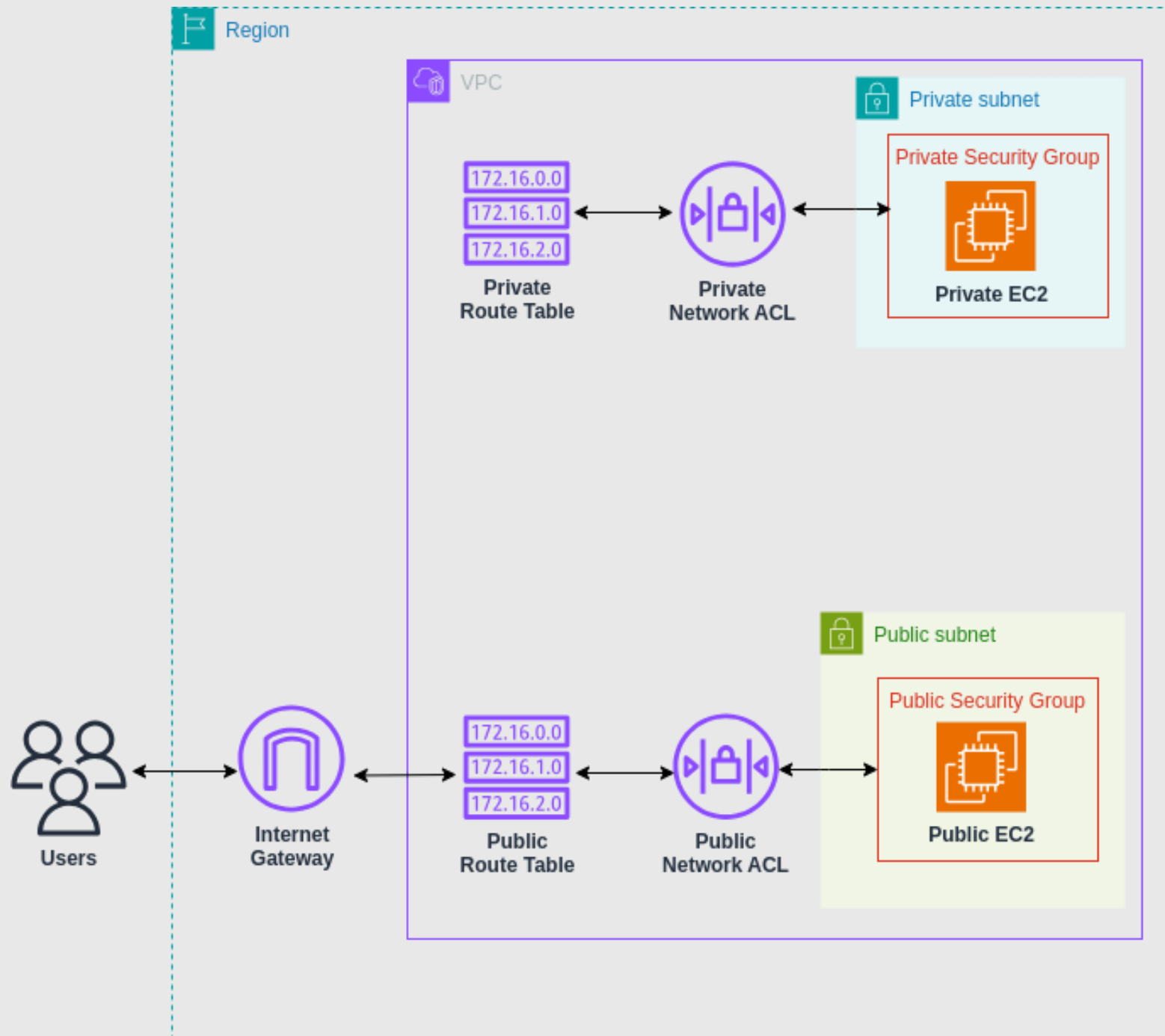
Subnets - Separate the VPC into public and private zones for organizing resource access.

Route Tables - Define how traffic flows between subnets and to the internet or other AWS services.

Internet Gateway - Allows public subnet instances to connect to the internet.

Amazon VPC Wizard - Quickly sets up VPC components using a guided, visual interface.

Architecture Diagram



The architecture has a VPC with a public subnet for an internet facing EC2 instance and a private subnet for a secure, internal EC2 instance. An internet gateway connects only the public subnet, and the VPC wizard sets up all components with a resource map

Steps to Implement

1. Launch the VPC Wizard:

Go to the VPC console and choose “VPC and more”. This guided wizard helps you quickly set up a VPC with both public and private subnets, route tables, and an internet gateway using a visual resource map.

2. Create EC2 in Public Subnet:

Launch an EC2 instance in the public subnet created by the wizard. Enable auto assign public IP and ensure the internet gateway is attached, so the instance can connect to the internet.

3. Create EC2 in Private Subnet:

Launch another EC2 instance in the private subnet, which doesn't have direct internet access. This instance is ideal for backend services or internal processing that require higher security.

4. Verify Connectivity:

Test internet access from the public instance and ensure the private instance remains unreachable from the internet. You can later set up a bastion host or NAT for secure, limited access if needed.

Story Time

VPC Wizard (Your Instant City Builder):

Head to your VPC console and choose “VPC and more” it’s like calling in an expert city planner who builds roads, zones, and gates in seconds! The visual map shows all your network parts coming together like magic.

Public Subnet Home (With Wi-Fi!):

Launch an EC2 instance here with a public IP, it’s like placing a smart house right at the city entrance, fully connected to the internet via the internet gateway. Great for hosting, updates, or acting as a gatekeeper.

Private Subnet Bunker (No Outsiders Allowed):

Launch another EC2 instance in the private zone, your secret lab! No internet, no noise, just internal only operations. Ideal for things like databases or backend servers.

Test the Setup (Check City Rules):

Make sure your public house can go online, while the private bunker stays hidden. It’s like ensuring only certain homes can send postcards, while others work quietly behind the scenes.

BY
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THANK YOU!

